

Machine Learning - Lab 2 Classification Models

Group 13

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1. COMPARING BOTH MODELS USED

K-Nearest Neighbors (KNN) and Decision Tree (DT) were the models chosen for this classification lab. Both models attempt to predict the popularity class of a music track using its musical characteristics, but they use different approaches.

KNN classifies a track by comparing it with similar tracks in the training data. It is relatively simple, but it requires feature scaling because the input variables have different numerical ranges. DT uses decision rules to separate the tracks into different classes. It can capture nonlinear relationships and does not require feature scaling.

The purpose of comparing these two models is to determine which approach performs better for this specific dataset, considering both the cross-validation results and the final evaluation on the test set.

K-Nearest Neighbors (KNN)

- **Strengths:** Simple to understand and implement. Can identify nonlinear relationships by comparing observations with nearby examples.
- **Weaknesses:** Requires feature scaling and its predictions depend on the number of neighbors selected. Predictions can become slower when working with large datasets.

Decision Tree (DT)

- **Strengths:** Can represent nonlinear relationships through decision rules and does not require feature scaling. Its structure is relatively easy to interpret, as the classification process follows a series of understandable decisions.
- **Weaknesses:** Can overfit the training data if the tree becomes too complex. It can also be sensitive to small changes in the training data, which may modify the tree structure.

As shown in the table obtained from the code, both models were evaluated using five-fold cross-validation before and after hyperparameter tuning.

Model	Baseline CV Accuracy	Tuned CV Accuracy	Improvement
K-Nearest Neighbors	0.379	0.381	0.002
Decision Tree	0.344	0.410	0.067

DT achieved the highest cross-validation accuracy after tuning. Its performance improved considerably when the model parameters were adjusted. KNN, on the other hand, improved only slightly.

However, the cross-validation results do not tell the whole story. The final evaluation on the test set produced a different ranking, which makes it necessary to compare the models using data that was not used for training or parameter selection.

2. FINAL EVALUATION ON THE TEST SET

Model	Precision	Recall	F1-score
KNN	39,4%	39,2%	39,3%
DT	32,8%	32,5%	32,5%

Accuracy: measures the proportion of all tracks classified correctly. KNN achieved a higher value, meaning that it made more correct predictions overall.

Precision: measures how often the model is correct when it predicts a particular class. The weighted Precision gives more importance to classes containing more tracks.

Recall: measures the proportion of tracks belonging to a class that the model identifies correctly. The weighted Recall is calculated across the three classes, taking their sizes into account.

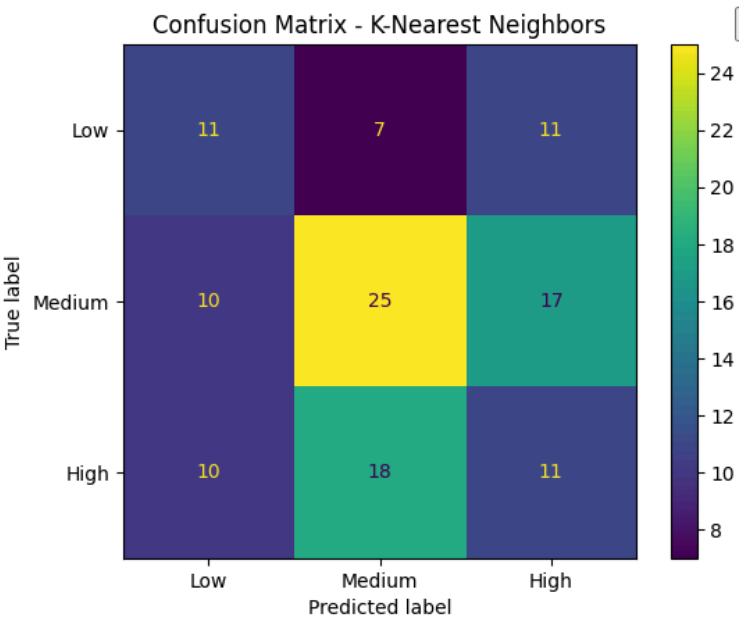
F1-score: combines Precision and Recall into a single metric. KNN achieved a higher weighted F1-score, indicating better overall performance when considering both measures.

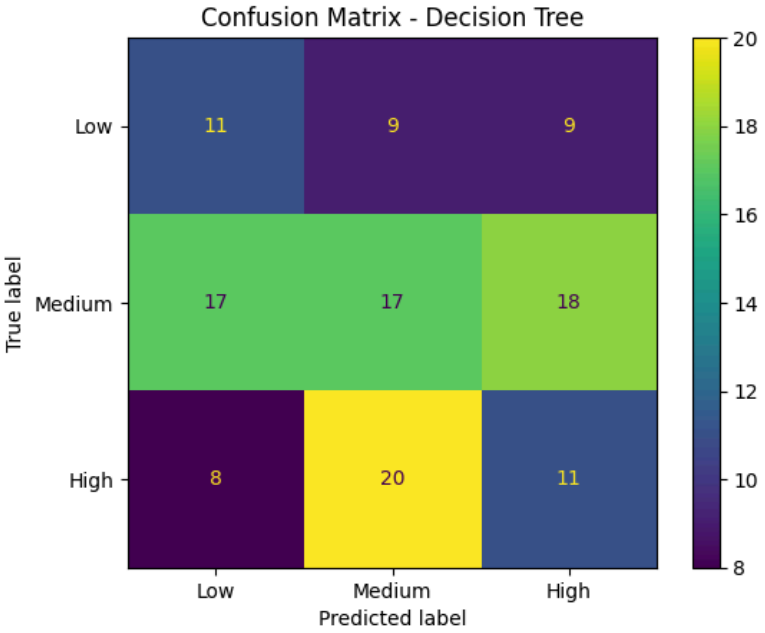
As shown by the results, KNN outperforms DT across all four test metrics. Nevertheless, neither model achieves particularly strong results, so this difference should not be interpreted as evidence that KNN is a highly accurate predictor.

Class Prediction Performance

The confusion matrices provide a more detailed understanding of the results by showing which classes are predicted correctly and which are more difficult to identify.

KNN correctly classified 25 out of 52 Medium tracks (48.1%), making this its best-predicted class. For the Low class, it correctly classified 11 out of 29 tracks (37.9%). Its performance was weakest for High, where it correctly identified 11 out of 39 tracks (28.2%).





DT correctly classified 11 out of 29 Low tracks (37.9%), 17 out of 52 Medium tracks (32.7%) and 11 out of 39 High tracks (28.2%). Therefore, its best result by class was for Low, while Medium and High were more difficult to predict.

Both models struggled to identify highly popular tracks. KNN incorrectly classified 18 High tracks as Medium, while DT made the same error with 20 tracks. This suggests that the selected musical characteristics do not clearly distinguish many highly popular tracks from those in the Medium class.

Overall, KNN performs better at identifying Medium tracks, while neither model demonstrates strong performance for High tracks. The confusion matrices therefore support the conclusion that the models have difficulty separating the popularity classes reliably.

A simple baseline model was also evaluated by always predicting the most frequent class: Medium: This model achieved a test accuracy of approximately 43,3% which is higher than both KNN (39,9%) and DT (32,5%).

Therefore although KNN performed better than DT, neither classifier improved on this simple baseline in terms of general accuracy. This indicates that both of the models have important limitations when predicting track popularity.

Recommended Model

For this dataset, KNN is the recommended model of the two evaluated because it achieves higher Accuracy, Precision, Recall and weighted F1-score on the test set. It also identifies the Medium class more successfully than DT.

However, the recommendation must be considered in context. DT achieved higher cross-validation accuracy after tuning, and both models performed worse than the simple baseline that always predicts Medium.

Consequently, KNN is the better of the two tested classifiers based on the final test metrics, but its predictive performance remains limited.